

LAMAP MALEKOULA ISLAND, Vanuatu

WMO: 915550

Lat: 16.422S

Lon: 167.809E

Elev: 85

StdP: 14.65

Time Zone: 11.00 (E11)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	67.8	69.5	57.7	71.4	72.9	59.8	77.1	73.1	23.5	76.0	21.4	76.8	4.0	120	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	88.0	80.6	87.2	80.2	86.5	79.7	82.2	86.7	81.5	86.0	80.9	85.4	7.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.7	85.8	80.1	156.6	85.0	79.5	153.6	84.4	46.1	87.0	45.3	86.3	44.7	85.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	20.4	18.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.2	81.4	81.5	81.4	80.4	79.0	77.5	76.6	76.1	77.4	78.6	79.7	80.7
	DBStd	2.72	1.65	1.72	1.55	1.57	1.70	2.05	2.33	2.64	2.41	2.13	1.85	1.73
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10651	973	883	972	913	899	825	824	809	823	886	892	952
	CDD65	5176	508	463	507	463	434	375	359	344	373	421	442	487
	CDH74	44495	5351	4919	5242	4384	3565	2494	2032	1837	2466	3347	3982	4876
(13)	CDH80	8874	1505	1426	1414	884	435	195	131	142	260	485	750	1247
(14)	Wind	WSAvg	8.0	6.6	6.4	6.2	7.7	8.8	8.5	9.2	9.2	9.3	8.2	7.1
(15) Precipitation	PrecAvg	75.7	9.3	8.5	10.9	8.5	7.2	4.5	4.1	3.6	4.2	4.8	6.4	5.4
	PrecMax	125.0	19.2	28.5	24.1	19.3	18.0	16.3	13.3	16.5	11.2	13.9	15.8	13.2
	PrecMin	48.7	1.1	0.8	4.4	4.4	0.4	0.7	1.2	0.4	0.2	0.2	1.0	1.4
	PrecStd	18.9	4.7	5.5	5.0	3.1	4.6	3.3	2.6	3.4	3.2	3.9	4.0	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.8	89.0	88.4	87.3	85.6	84.6	84.6	84.7	85.5	86.2	87.7	88.1
		MCWB	80.7	81.6	80.8	80.9	79.3	78.4	78.0	76.9	78.0	78.5	80.3	79.6
	2%	DB	87.6	87.9	87.5	86.3	84.6	83.2	82.8	82.8	84.1	84.9	86.1	87.1
		MCWB	80.5	80.7	80.7	80.2	78.6	77.1	76.3	75.9	77.2	77.5	79.0	79.2
	5%	DB	86.7	87.0	86.6	85.3	83.6	82.2	81.3	81.4	82.8	83.7	85.0	86.2
		MCWB	79.8	80.1	80.2	79.5	77.9	76.4	75.1	74.7	76.0	76.7	78.3	79.0
	10%	DB	85.8	86.0	85.6	84.2	82.4	81.1	80.1	80.1	81.6	82.7	84.0	85.3
		MCWB	79.2	79.5	79.7	78.8	77.0	75.7	74.3	73.7	75.1	76.0	77.6	78.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.2	83.1	82.6	82.2	81.2	79.9	79.2	78.2	79.6	80.1	81.8	82.2
		MCDB	87.3	87.8	87.1	86.5	84.6	83.0	83.0	82.6	84.3	84.6	86.3	86.7
	2%	WB	81.9	81.9	81.6	81.0	79.7	78.6	77.7	77.1	78.3	79.0	80.3	81.1
		MCDB	86.2	86.8	86.2	85.2	83.4	81.5	81.2	81.4	82.9	83.5	84.7	85.8
	5%	WB	81.1	81.0	80.9	80.2	78.8	77.7	76.6	76.2	77.4	78.0	79.3	80.3
		MCDB	85.3	85.9	85.5	84.3	82.5	80.8	79.9	80.2	81.6	82.4	83.7	85.0
	10%	WB	80.4	80.3	80.2	79.5	78.0	76.9	75.7	75.3	76.4	77.2	78.4	79.4
		MCDB	84.5	84.9	84.6	83.3	81.5	80.2	79.2	79.1	80.4	81.4	82.8	84.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.8	8.6	7.7	6.7	6.9	6.9	7.7	7.7	7.8	8.1	8.7
		MCDBR	9.9	9.9	9.7	8.9	8.2	8.2	8.3	9.1	9.1	9.0	9.6	10.1
	5% WB	MCWBR	5.1	5.1	5.2	4.9	4.8	4.6	4.8	5.0	5.1	5.1	5.2	5.2
		MCWBR	8.5	9.4	8.9	8.3	7.9	6.9	7.1	8.1	8.2	8.0	9.0	9.1
(40) Clear-Sky Solar Irradiance	taub		0.408	0.400	0.393	0.387	0.394	0.411	0.399	0.393	0.405	0.406	0.410	0.400
		taud	2.533	2.565	2.592	2.610	2.559	2.492	2.516	2.530	2.484	2.502	2.505	2.540
	Ebn at Noon		297	297	292	283	269	258	265	277	284	292	295	299
		Edh at Noon	36	34	32	30	30	30	30	32	35	36	36	35
(44) All-Sky Solar Radiation	RadAvg		1941	1875	1736	1486	1272	1127	1183	1381	1644	1782	1889	2016
	RadStd		184	151	145	134	125	88	81	98	149	168	175	137

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page